The Observer’s Sphere Model

Attention, Nested Worldviews, and the Etched Landscape of Forms

A geometric account of how biology, experience, and attention construct an operational picture of reality, with nested shells whose inner etchings modulate the light available on outer ones.

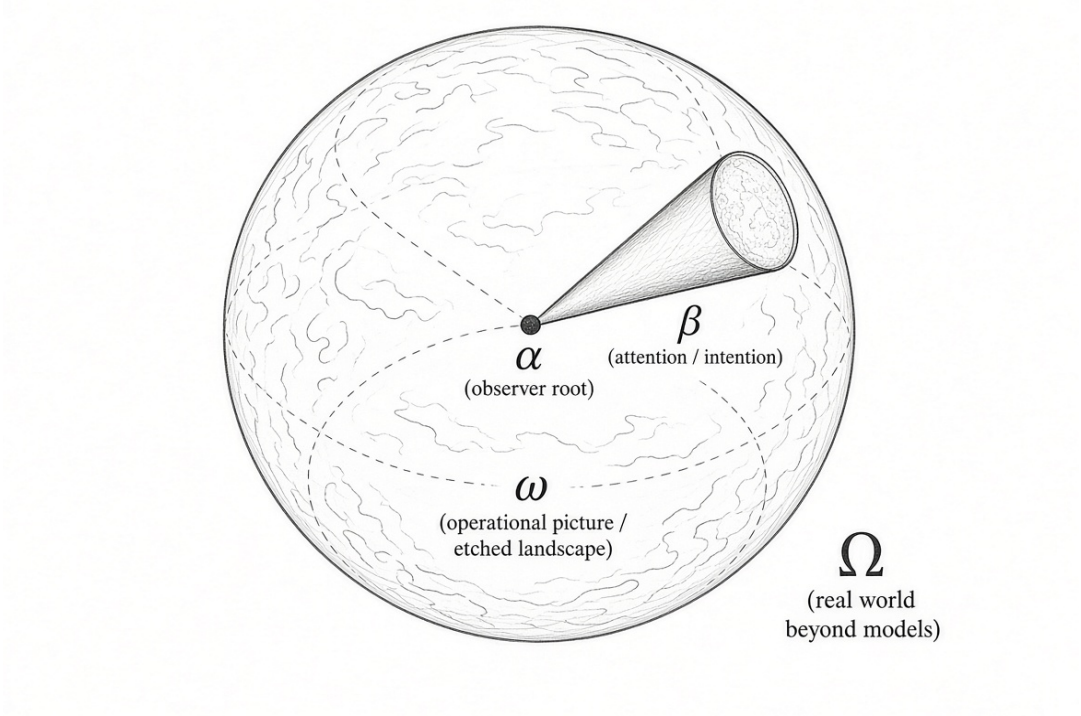


Figure 1. Basic geometric structure: α at the center, β the attentional cone, ω the etched interior, Ω the exterior.

1. Core Components

Every human being carries a worldview as a condition of being. Biology installs a representation of how things look and feel; this representation can be incomplete or wrong, and it is revisable by new data. The model captures this situation geometrically.

α — Observer Root

The central point of the system. The locus from which attention originates and the light of observation is cast.

ω — Operational Picture

The interior of the (outer) sphere. The biologically generated, experience-etched landscape of forms that constitutes the observer’s working model of the world. In the nested extension, ω is the outermost shell.

β — Cone of Attention

The solid cone extending from α outward. Its direction and aperture determine which portion of any shell is actively illuminated. β can rotate and can cross shell boundaries.

Ω — Exterior Reality

The real world beyond models and senses. Biologically unknowable in full. The boundary between Ω and the outermost shell is not fixed.

2. Formalization

Let a sphere have radius R. The attentional cone β subtends a solid angle at α.

$$\Omega_{attn} = 2\pi\ (1 - \cos\ \phi)$$
$$A = 2\pi\ R^2\ (1 - \cos\ \phi)$$

A measures momentary observational capacity on a given shell. Boundary radius may fluctuate: $R(t) = R_0 + \Delta R$(experience, instruments, concepts).

3. Nested Shells

The operational picture can be resolved into stacked or nested shells. Etchings on an inner shell filter and shape the light that reaches every outer shell, so that foundational structure systematically biases which patterns remain available farther out.

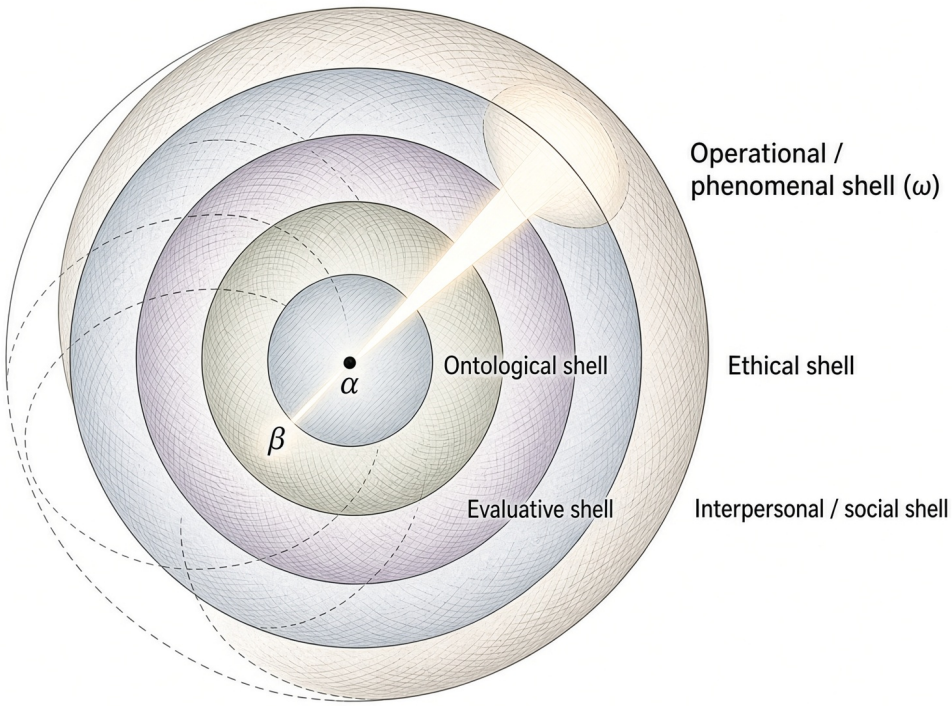


Figure 2. Nested shells. Light from α is filtered by each successive shell before reaching the outer operational surface.

Proposed ordering (innermost to outermost)

Ontological shell. Core etchings around “why there is something rather than nothing.” Implicit or explicit answers set the basic character of the light transmitted outward.

Evaluative / axiological shell. Patterns of what counts as real, good, or significant once existence is given. Contingency versus necessity, intrinsic value versus indifference.

Ethical shell. Concrete moral etchings (harm, fairness, authority, purity, loyalty, etc.). Relativist versus absolutist families of structure are differentially illuminated according to the light arriving from within. Specific content still determines direction.

Interpersonal / social shell. Etchings of attachment, status, reciprocity, conflict, and care. These inherit the ethical illumination.

Operational / phenomenal shell (outer ω). The everyday landscape of forms through which the observer actually attends. All prior shells modulate which regions are easy for β to illuminate.

Changes on an inner shell propagate outward by altering the relative brightness and accessibility of families of etchings on subsequent shells. The reverse influence is weaker but possible over time.

4. Developmental Etching and Archetypal Structure

As the observer grows, shells are installed through experience and play. The inner surfaces are etched by language, story, and the internalization of repeated patterns. The earliest etchings include archetypal forms; later forms are layered on top or grown into them in a fractal-like manner.

When the light of the observer passes over these etchings, they generate the internal mental representations of the observed subject. True change of mind requires re-etching; merely redirecting β changes which region is illuminated but does not rewrite the structure.

$$E(s,\ t) = E_{\text{archetype}}(s) + \int L(s,\ \tau)\ d\tau$$

5. States of the Attentional Cone

I-state (internal): β rotates freely at the will of the observer within or across shells.

x-state (present): β is locked to ongoing perception. Its direction correlates with where the observer is actually looking and perceiving.

6. Spinning and Thought Motion

Successive thoughts arise when illumination of one etching pulls β toward related ones. Speed and path geometry distinguish patterns people report as spinning:

• Slow closed loops on densely etched negative regions $\rightarrow$ classic rumination.
• Fast closed loops $\rightarrow$ urgent anxious spirals.
• Fast open trajectories across distant regions $\rightarrow$ racing or jumping thoughts.
Domain shells make these patterns sensitive to content: circular high-recurrence spin is more characteristic of interpersonal, moral, and self-related shells; faster variable trajectories appear more often on physical or instrumental regions.

7. Implications

All observation is of the model. Sensory data are conformed through the shells before they populate or revise the forms. Inner etchings (especially ontological ones) change the light under which outer domains—ethics, social relations, everyday experience—are seen. Scientific instruments and conceptual advances can expand or refine shells and shift the boundary with Ω, but they do not exit the nested structure.

The model remains a compact geometric picture of phenomena studied under other formalisms (internal generative models, attentional selection, developmental schemas, hierarchical worldviews). The particular combination of nested shells, central root, rotatable cone, and etched surfaces is a personal synthesis.

— End of illustration —